Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “recoil” and “external” to explain your model or result. Sentence starters are optional.

recoil: motion backward when another part is sent forward

external: acting from outside the chosen system

conserved: unchanged in total under the stated conditions

01 Bring back an earlier idea

A cart has momentum -15 kg·m/s and mass 3 kg. **Find** velocity and its new momentum if velocity reverses at unchanged speed.

Word help: momentum = mass times velocity, including direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part’s momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two carts push apart from rest on an ideal track. **Draw** their momentum arrows. How must the two momenta compare?

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

04 Solve and justify

A 60 kg person and 3 kg tool are initially at rest relative to a spacecraft. The tool leaves at +8 m/s in that frame. **Find** person velocity, neglecting external impulse.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A projectile's velocity is (12, -16) m/s.
Find its speed. What information would locate it?

Word help: projectile = an object moving after launch under the stated forces

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Two fragments of masses 2 and 3 kg separate from rest. The 2 kg fragment moves at +9 m/s.
Find the other velocity and total kinetic energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 60 kg person and 10 kg board start at rest. After a push, the board moves at +3 m/s relative to the ground.
Find the person's ground velocity and the board's speed relative to the person.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3